

Positivity 3.1

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1 Morse Theory

In this section, let M be a real smooth manifold of dimension m , and $f : M \rightarrow \mathbb{R}$ be a smooth function.

Definition 1.1. A smooth function $f : M \rightarrow \mathbb{R}$ is called a exhaustion function if for any $a \in \mathbb{R}$, $M^a := f^{-1}(-\infty, a]$ is compact.

Definition 1.2. A point $p \in M$ is called a critical point of a smooth function $f : M \rightarrow \mathbb{R}$ if df_p is not surjective. When $N = \mathbb{R}$, this is equivalent to $\nabla f_p = 0$. The set of critical points is denoted by $\text{Crit}(f)$.

Definition 1.3. A critical point $p \in M$ of a smooth function $f : M \rightarrow \mathbb{R}$ is called non-degenerate if $\text{Hess}_p(f)$ is non-degenerate.

Definition 1.4. A smooth function $f : M \rightarrow \mathbb{R}$ is called Morse function if every critical points are non-degenerate.

Definition 1.5. (Relative CW complex) Let X be a topological space and A a subset of X , then (X, A) is called a relative CW complex if there exists

$$A = X^{(-1)} \subseteq X^{(0)} \subseteq \dots \subseteq X^{(n)} \subseteq \dots \subseteq X$$

such that

1. For every n , there exists an index set J_n and maps

$$\varphi_j^n : S^{n-1} \rightarrow X^{(n-1)} \quad \forall j \in J_n$$

such that $X^{(n)} = X^{(n-1)} \amalg \amalg_j D^n / \sim$, where $x \sim \varphi_j^n(x)$ for $x \in S^{n-1}$

2. $X = \bigcup_{n \geq 0} X^{(n)}$

3. X has weak topology i.e. $U \subseteq X$ is open iff $U \cap X^{(n)}$ is open for all $n \geq 0$.

Proposition 1.1. Let $f : M \rightarrow \mathbb{R}$ be a Morse function, then $\text{Crit}(f)$ is discrete.

Proof. Given $p \in \text{Crit}(f)$, we pick a local chart (φ, U, V) such that $p \in U$ and $\varphi(U) = V \subseteq \mathbb{R}^m$ with local coordinate $x_1, \dots, x_m$. Let $g = f \circ \varphi^{-1}$, then one can define

$$F : V \longrightarrow \mathbb{R}^m$$
$$x \longmapsto \left(\frac{\partial g}{\partial x_1}, \dots, \frac{\partial g}{\partial x_m} \right)$$

By definition $x \in \text{Crit}(f)$ iff $\varphi(x) \in F^{-1}(0)$. Thus $\varphi(p) \in F^{-1}(0)$. On the other hand, $(dF)_{\varphi(p)} = \text{Hess}_{\varphi(p)}(g)$ is non-degenerated since f is a Morse function and φ is a diffeomorphism. By inverse function theorem, there exists open subsets $\varphi(p) \in W \subseteq V$ and $0 \in W' \subseteq \mathbb{R}^m$ such that $F|_W : W \rightarrow W'$ is a diffeomorphism. In particular, F is injective, which says that on W there is only one points i.e. $\varphi(p)$ such that $F(\varphi(p)) = 0$. Thus $\varphi(p)$ is the only critical point on W and we are done. $\square$

Theorem 1.2. Let $f : M \rightarrow \mathbb{R}$ be a smooth exhaustion Morse function. Then M has the homotopy type of a CW complex, where cells one-to-one correspond to critical points such that the dimension of cell is just $\text{Index}_p(f)$. Concretely, (M, M^a) is a relative CW complex, where a is a regular value of f , such that

$$M = M^a \cup \bigcup_{p \in \text{Crit}(f|_{M-M^a})} e_p^{\text{Index}_p(f)}$$

Proof. See Milnor's Morse Theory. $\square$

Theorem 1.3. Let $M \subseteq \mathbb{R}^N$ be a closed smooth submanifold. Fix a point $c \in \mathbb{R}^N$ and let

$$\begin{aligned} \phi_c : M &\longrightarrow \mathbb{R} \\ x &\longmapsto \|x - c\|^2 \end{aligned}$$

where $\|\cdot\|$ is the Euclidean norm. Then for general $c \in \mathbb{R}^N$ (i.e. except a set of measure zero set), ϕ_c is exhausting Morse function.

Proof. First we prove M^a is compact. Note that $M^a = f^{-1}(-\infty, a] = f^{-1}[0, a] \subseteq B(0, \|c\| + a)$. The compactness of $[0, a]$ says that it is closed and the continuous of f says that M^a is closed, which is contained in a compact set and thus compact also. For the second part, given any $x \in M$ and $u \in T_x M$,

$$(d\phi_c)_x(u) = 2 \langle x - c, u \rangle$$

Thus $x \in \text{Crit}(\phi_c)$ iff $\langle x - c, u \rangle = 0$ for every $u \in T_x M$ i.e. $x - c \in N_x M$. Hence if we consider the map (here $NM = \{(x, v) \mid x \in M, v \in N_x M\}$ is the normal bundle)

$$\begin{aligned} E : NM &\longrightarrow \mathbb{R}^N \\ (x, v) &\longmapsto x + v \end{aligned}$$

then $x \in \text{Crit}(\phi_c)$ iff $(x, c - x) \in E^{-1}(c)$. We claim that $x \in \text{Crit}(\phi_c)$ is non-degenerate iff $(x, c - x)$ is a regular point of E (i.e. not a critical point). Then

$$\begin{aligned} &\{c \in \mathbb{R}^N \mid \phi_c \text{ is not Morse function}\} \\ &= \{(x, c - x) \in NM \mid (x, c - x) \text{ is the critical point of } E\} \\ &\subseteq \{c \in \mathbb{R}^N \mid c \text{ is the critical value of } E\} =: S \end{aligned}$$

Use Sard's theorem, we know that S is a measure zero set, i.e. for almost every c , ϕ_c is a Morse function. $\square$

Theorem 1.4. (Sard's theorem) For any smooth map $f : M \rightarrow N$, $f(\text{Crit}(f))$ is a measure zero set in N .

2 Some Cohomology

In this section, let X be a topological space, $Z \subseteq X$ be a closed subspace and $\mathcal{F}$ be a sheaf of abelian groups on X . We define

$$H^0(Z, \mathcal{F}) = \varinjlim_{\substack{Z \subseteq U \\ U \text{ open}}} H^0(U, \mathcal{F})$$

Definition 2.1. (Paracompact) X is called paracompact if for every open covering $\{U_i\}_{i \in I}$ of X there exists a locally finite refinement i.e. there exists a open covering $\{V_j\}_{j \in J}$ of X such that

1. For every $j \in J$ there exists $i \in I$ such that $V_j \subseteq U_i$.
2. For every point $x \in X$, there exists an open neighborhood W of x such that $\{j \in J : V_j \cap W \neq \emptyset\}$ is finite.

Definition 2.2. (Soft sheaf) $\mathcal{F}$ is called soft sheaf if for every closed subset $E \subseteq X$, the restriction

$$H^0(X, \mathcal{F}) \rightarrow H^0(E, \mathcal{F})$$

is surjective.

Proposition 2.1. Any flasque sheaf $\mathcal{F}$ is a soft sheaf.

Proof. Given any element $s \in H^0(Z, \mathcal{F})$, we choose an representation (s_U, U) . Now use $H^0(X, \mathcal{F}) \twoheadrightarrow H^0(U, \mathcal{F})$. □

Proposition 2.2. If X is paracompact Hausdorff (e.g. X is a real smooth manifold.) and given any short exact sequence of sheaves

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{G} \longrightarrow \mathcal{H} \longrightarrow 0$$

, the following holds.

1. If $\mathcal{F}$ is soft, then

$$0 \longrightarrow H^0(Z, \mathcal{F}) \longrightarrow H^0(Z, \mathcal{G}) \longrightarrow H^0(Z, \mathcal{H}) \longrightarrow 0$$

is exact for every closed subset $Z \subseteq X$.

2. If $\mathcal{F}, \mathcal{G}$ are soft then, so is $\mathcal{H}$.
3. If $\mathcal{F}, \mathcal{H}$ are soft then, so is $\mathcal{G}$.
4. Given any sheaf $\mathcal{F}$, we can get $0 \rightarrow \mathcal{F} \hookrightarrow \mathcal{I}$ for some soft sheaf $\mathcal{I}$.
5. If $\mathcal{F}$ is soft, then $H^i(X, \mathcal{F}) = 0$ for all $i > 0$.

Proof. Same as the proof of flasque sheaf. Paracompact Hausdorff property is used to ensure existence of the extension of sections/glueing of sections. □

Theorem 2.3. For paracompact Hausdorff space, we have

$$H^i(Z, \mathcal{F}|_Z) = \varinjlim_{\substack{Z \subseteq U \\ U \text{ open}}} H^i(U, \mathcal{F})$$

Proof. We pick a flasque sheaf resolution

$$0 \longrightarrow \mathcal{F} \longrightarrow \mathcal{I}^0 \longrightarrow \mathcal{I}^1 \longrightarrow \dots$$

By applying $H^0(U, -)$ (note that the restriction of flasque to open subset is flasque) and take cohomology, one can find

$$\lim_{\substack{\longrightarrow \\ Z \subseteq U}} H^i(U, \mathcal{F}) = \lim_{\substack{\longrightarrow \\ Z \subseteq U}} h^i(C^\bullet(U, \mathcal{F})) = h^i(\lim_{\substack{\longrightarrow \\ Z \subseteq U}} C^\bullet(H^0(U, \mathcal{F}))) = h^i(C^\bullet H^0(Z, \mathcal{F}|_Z)) = H^i(Z, \mathcal{F}|_Z)$$

The final equality holds since the restriction of soft sheaf into closed subset is soft and the Z is also paracompact Hausdorff, which implies $H^i(Y, \mathcal{I}^i|_Y) = 0$ for $i > 0$. $\square$

Theorem 2.4. Let $(A_i, f_{ii'})_{i \in I}$ and $(B_j, g_{jj'})_{j \in J}$ be two direct systems, if there exists order-preserving $\phi : I \rightarrow J$ such that for every $j \in J$ there exists $i \in I$ such that $j \leq \phi(i)$ and $\eta_i : A_i \cong B_{\phi(i)}$ such that $\eta_{i'} \circ f_{ii'} = g_{\phi(i)\phi(i')} \circ \eta_i$. Then

$$\lim_{\substack{\longrightarrow \\ i \in I}} A_i \cong \lim_{\substack{\longrightarrow \\ j \in J}} B_j$$

Proof. Directly check. $\square$

Proposition 2.5. If $f : Y^{an} \rightarrow X^{an}$ is a finite morphism of complex algebraic varieties and $\mathcal{F} \in \mathbf{Sh}(Y^{an})$, then

$$(f_* \mathcal{F})_x = \bigoplus_{y \in f^{-1}(x)} \mathcal{F}_y$$

Proof. First we know that f is closed map since X^{an} is locally compact Hausdorff and the fiber is finite. Given $x \in X$ and $f^{-1}(x) = \{y_1, \dots, y_d\}$ and any open neighborhood W . By Hausdorff property and first countable property, we can choose disjoint neighborhood $W_i \ni y_i$ such that $W_i \subseteq f^{-1}(W)$. Let $U = W - f(Y^{an} - \bigcup_{i=1}^d W_i)$, then $f^{-1}(U)$ is the union of disjoint open subsets. Thus

$$(f_* \mathcal{F})(U) = \mathcal{F}(f^{-1}(U)) = \bigoplus_{i=1}^d \mathcal{F}(U_i)$$

Now take direct limits on both sides. $\square$

Theorem 2.6. (Leray spectral sequence) Let $f : Y \rightarrow X$ be a morphism, then we have

$$E_2^{pq} = H^q(X, R^p f_* \mathcal{F}) \Rightarrow H^{p+q}(Y, \mathcal{F})$$

Proposition 2.7. If X is a locally contractible topological space and A any abelian group, then we have

$$H_{\text{Sing}}^i(X; A) = H^i(X; \underline{A}_X)$$

for every $i \geq 0$. In particular, this is true for every complex variety with Euclidean topology by 3.2.

Proposition 2.8. (Poincare duality) Let M be a orientable real smooth manifold of dimension m without boundary, then we have isomorphism

$$H_c^q(M; \mathbb{Z}) \xrightarrow{\cap [M]_{B^M}} H_{m-q}(M; \mathbb{Z})$$

Proof. See any book of algebraic topology/differential geometry. $\square$

Proposition 2.9. Let X be a locally compact Hausdorff, locally contractible space. For any closed subsets $A \subseteq X$, there is an isomorphism

$$H^q(X, A; \mathbb{Z}) = H_c^q(X - A; \mathbb{Z})$$

Proof. Compare the long exact sequence of compact support cohomology and long exact sequence of the pair of spaces and use five lemma. $\square$

3 Homotopy Theory

Definition 3.1. (Semi-algebraic set) A subset $A \subseteq \mathbb{R}^n$ is called a semi-algebraic set if there exists finite polynomials $f_{ij}, g_{ij} \in \mathbb{R}[x_1, \dots, x_n]$, $(i \in I, j \in I_j)$ such that

$$A = \bigcup_{i \in I} \{x \in \mathbb{R}^n \mid f_{ij}(x) = 0, g_{ij} > 0, \forall j \in J_i\}$$

Proposition 3.1. The finite union/intersection and difference of semi-algebraic set is also semi-algebraic. In particular, if $A \subseteq X$ is subvariety of real/complex variety X , then $A, X, X - A$ are all semi-algebraic.

Proof. Directly check. □

Theorem 3.2. Let $X \subseteq \mathbb{R}^n$ be a compact semi-algebraic set and $A_1, \dots, A_m \subseteq X$ be semialgebra subset, then there exists a finite simplicial complex and a semi-algebraic homeomorphism

$$h : |K| \rightarrow X$$

such that there exists $S_i \subseteq K$ such that $A_i = h(\bigcup_{\sigma \in S_i} \text{Int}(\sigma))$.

Remark 3.1. The theorem is want to say that there exists a compatible triangulated structure on $(X, A_1, \dots, A_m)$.

Corollary 3.3. Given any compact affine complex/real variety X , and algebraic subsets $A_1, \dots, A_n$ of X , then any A_i is locally contractible and there exists open neighborhood of A_i which deforms to A_i .

The following part can consult Hatcher chapter 4. Here we list some definition and property without proof.

Definition 3.2. (Homotopy groups) let X be a topological space, $A \subseteq X$ subspaces, $I^i = [0, 1]^i$, $J^{i-1} = \overline{\partial I^i - \{0\}} \times [0, 1]^{i-1}$ and $x_0 \in A$. Then we define the relative homotopy group with base point x_0 to be

$$\pi_i(X, A, x_0) = \{f : (I^i, \partial I^i, J^{i-1}) \rightarrow (X, A, x_0) \mid f(\partial I^i) \subseteq A, f(J^{i-1}) = x_0\} / \sim$$

where the equivalent relationship is homotopy equivalence i.e. $f \sim g$ iff there exists $H : I^i \times I \rightarrow X$ such that

1. $H(\partial I^i, t) \subseteq A$.
2. $H(J^{i-1}, t) = x_0$.
3. $H(x, 0) = f$.
4. $H(x, 1) = g$.

If $\varphi : (X, A, x_0) \rightarrow (Y, B, y_0)$ is a continuous map such that $f(A) \subseteq B$ and $f(x_0) = y_0$, then it induces

$$\begin{aligned} \varphi_* : \pi_i(X, A, x_0) &\longrightarrow \pi(Y, B, y_0) \\ f &\longmapsto \varphi \circ f \end{aligned}$$

If $A = \{x_0\}$ and $B = \{y_0\}$, we get the absolute case of homotopy groups.

Remark 3.2. $\pi_i(X, A)$ is only a set when $i = 1$, a group when $i \geq 2$ and an abelian group when $i \geq 3$.

Definition 3.3. Recall that X is path-connected if $\pi_0(X, x_0) = \{\text{pt}\}$ and is simply-connected if $\pi_1(X, x_0) = 0$ and $\pi_0(X, x_0) = \{\text{pt}\}$.

Definition 3.4. A continuous map $f : (X, x_0) \rightarrow (Y, y_0)$ is called r -connected if $f_* : \pi_i(X, x_0) \rightarrow \pi_i(Y, y_0)$ is isomorphism for $i < r$ and surjective for $i = r$.

Proposition 3.4. π_i is a functor.

Proposition 3.5. If $i : A \rightarrow X$ is a homotopy equivalence, then

$$\pi_i(X, A, x_0) = 0$$

for all $i \geq 1$ and $\pi_0(X, A, x_0) = \{\text{pt}\}$.

Proposition 3.6. We give two long exact sequence.

1. Given inclusion $i : A \subseteq X$, we have a long exact sequence of homotopy groups

$$[f] \longmapsto [f|_{\partial I^i}]$$

$$\pi_i(A, x_0) \xrightarrow{i_*} \pi_i(X, x_0) \longrightarrow \pi_i(X, A, x_0) \xrightarrow{\partial} \pi_{i-1}(A, x_0)$$

$$[f] \longmapsto [f]$$

2. Let $x_0 \in B \subseteq A \subseteq X$, we have a long exact sequence of homotopy groups

$$\pi_i(A, B) \longrightarrow \pi_i(X, B) \longrightarrow \pi_i(X, A) \longrightarrow \pi_{i-1}(A, B)$$

Proposition 3.7. If $X = A \cup \bigcup_{\alpha} e_{\alpha}^{d_{\alpha}}$ (i.e. a relative CW complex) with $d_{\alpha} \geq n$, then

$$\pi_i(X, A) = 0$$

for all $i < n$.

4 Complex Geometry

Theorem 4.1. (GAGA) Let X be a complex projective variety with X^{an} its corresponding analytic space. Then

1. The category of coherent sheaves on X is equivalent to the category of coherent sheaves on X^{an}
2. For every $i \geq 0$, we have

$$H^i(X, \mathcal{F}) \cong H^i(X^{an}, \mathcal{F}^{an})$$

Proposition 4.2. Let X be a complex projective variety. Then

$$\text{Pic}(X^{an}) = \text{Pic}(X)$$

Proof. Direct conclusion of 4.1. □

Theorem 4.3. (Hodge decomposition) For compact Kahler manifold X (e.g. smooth complex projective variety), we have Hodge decomposition

$$H^r(X, \mathbb{C}) = \bigoplus_{p+q=r} H^{p,q}(X) = \bigoplus_{p+q=r} H^q(X, \Omega_X^p)$$

5 Topology of Affine Variety

In this section, we will consider the Zariski and Euclidean topology/cohomology of affine varieties $X \subseteq \mathbb{C}^r$. If necessary, we use

$$H_{\text{Sing}}^i(X; \mathbb{Z}), H^i(X, \mathcal{F}), H^i(X^{an}, \mathcal{F})$$

to distinct the singular cohomology of X , sheaf cohomology of sheaf $\mathcal{F}$ w.r.t. Zariski topology of X and sheaf cohomology of $\mathcal{F}$ w.r.t. Euclidean topology of X .

Theorem 5.1. (Andreotti-Frankel) Let $V \subseteq \mathbb{C}^r$ be a closed connected complex submanifold of complex dimension n . Then V has the homotopy type of a CW complex of real dimension $\leq n$. Consequently, given every abelian group A , we have

$$H^i(V; A) = H_i(V; A) = 0 \quad \forall i > n$$

Proof. We can choose a Morse function ϕ_c by 1.3. Then by 1.2 we just need to show that $\text{Index}_p(\phi_c) \leq n$, which is given by the following proposition. $\square$

Proposition 5.2. (Index estimate) Let $V \subseteq \mathbb{C}^r$ be a closed complex submanifold of complex dimension n . Fix a general point $c \in \mathbb{C}^r = \mathbb{R}^{2r}$, and if $p \in V$ is any critical point of ϕ_c , then

$$\text{Index}_p(\phi_c) \leq n$$

Proof. Let $r = n + k$, WLOG, by transition and rotation we assume $p = 0, c = e_{n+1}$ and locally $V = \{(y, f(y)) \in \mathbb{C}^n \times \mathbb{C}^k\}$ such that $df_0 = 0$. Then

$$\begin{aligned} \phi_c(z) &= |z_1|^2 + |z_2|^2 + \cdots + |z_n|^2 + |f_1(z) - 1|^2 + |f_2(z)|^2 + \cdots + |f_k(z)|^2 \\ &= (1 - 2\Re f_1(z)) + \sum_{i=1}^n |z_i|^2 + \sum_{i=1}^k |f_i(z)|^2 \end{aligned}$$

Assume that $f_1(z) = 0 + 0 \cdots z + Q(z) + (\text{higher terms})$, then we know

$$\text{Hess}_0(\phi_c) = 2I - 2\text{Hess}_0(\Re(Q(z)))$$

If λ is an eigenvalue of $\text{Hess}_0(\Re(Q(z)))$, we know that $2 - 2\lambda$ is an eigenvalue of $\text{Hess}_0(\phi_c)$. So the number of negative eigenvalues is no more than the number of positive eigenvalues of $\text{Hess}_0(\Re(Q(z)))$, which is given by the following lemma. $\square$

Lemma 5.3. Let $Q(z)$ be a complex homogeneous form of degree 2. Then $\text{Hess}_0(\Re(Q(z)))$ has at most n positive eigenvalues and at most n negative eigenvalues.

Proof. Assume $z = (z_1, \dots, z_n)^T = x + iy, Q(z) = z^T(A + iB)z$ with A, B real symmetric matrice, then

$$\begin{aligned} Q(z) &= (x^T + iy^T)(A + iB)(x + iy) = (x^T Ax - y^T Bx - x^T By - y^T Ay) + i(x^T Bx + y^T Ax + x^T Ay - y^T By) \\ \Re(Q(z)) &= x^T Ax - y^T Bx - x^T By - y^T Ay = x^T Ax - 2x^T By - y^T Ay = x^T \begin{pmatrix} A & -B \\ -B & -A \end{pmatrix} y \end{aligned}$$

So if $\begin{pmatrix} u \\ v \end{pmatrix}$ is an eigenvector with eigenvalue λ , then $\begin{pmatrix} -v \\ u \end{pmatrix}$ is an eigenvector with eigenvalue $-\lambda$. So the number of positive eigenvalues is the same as that of negative eigenvalues. $\square$

Proposition 5.4. Let $V \subseteq \mathbb{C}^r$ as in the previous theorem, then $H_n(V; \mathbb{Z})$ is torsion-free.

Proof. We use universal coefficient theorem in topology

$$H_{n+1}(V; F) \cong (H_{n+1}(V; \mathbb{Z}) \otimes F) \oplus \text{Tor}_1^{\mathbb{Z}}(H_n(V; \mathbb{Z}), F)$$

From above theorem, we know that the CW-complex is bounded by n and thus $\text{Tor}_1^{\mathbb{Z}}(H_n(V; \mathbb{Z}), F) = 0$ for every abelian group F . And this says that $H_n(V; \mathbb{Z})$ is torsion-free. Otherwise $pa = 0$ for some $a \in H_n(V; \mathbb{Z})$ and some prime p , then applying $H_n(V; \mathbb{Z}) \otimes -$ to the exact sequence

$$0 \longrightarrow \mathbb{Z} \xrightarrow{\cdot p} \mathbb{Z} \longrightarrow \mathbb{F}_p \longrightarrow 0$$

and taking $\text{Tor}_1^{\mathbb{Z}}$ we find $\text{Tor}_1^{\mathbb{Z}}(H_n(V; \mathbb{Z}), \mathbb{F}_p) \neq 0$. □

We give two example that one has homotopy type of finite CW complex while the other not.

Example 5.1. Let $V \subseteq \mathbb{C}^2$ be the curve defined by $w \cdot \sin(\pi z) - 1 = 0$. Then under the map

$$V \longrightarrow \mathbb{C} - \mathbb{Z} \qquad V \longleftarrow \mathbb{C} - \mathbb{Z}$$

$$(z, w) \longmapsto z \qquad \left(z, \frac{1}{\sin(\pi z)}\right) \longmapsto z$$

V is isomorphic to $\mathbb{C} - \mathbb{Z}$, which is homotopy to $\vee_{\mathbb{Z}} S^1$. Hence $H_1(V; \mathbb{Z}) = \bigoplus_{n \in \mathbb{Z}} \mathbb{Z}$ is a infinite rank abelian group, which says that V do not have a homotopy type of finite CW complex.

Example 5.2. Let $V \subseteq \mathbb{C}^r$ be a smooth affine algebraic variety of complex dimension n . Then V has the finite homotopy type of a finite CW complex of real dimension $\leq n$.

Proof. We pick a general $c \in \mathbb{C}^r$, by 5.2, we know that it is a polynomial, and thus its critical locus $\text{Crit}(\phi_c)$, which is defined by polynomials, is real algebraic variety of V . Finally, one can show that the Krull dimension of $\text{Crit}(\phi_c)$ is 0. Otherwise we can consider the smooth locus $\text{Crit}(\phi_c)_{\text{sm}}$, which is a smooth real manifold of the same dimension as its Krull dimension. This is a contradiction since $\text{Crit}(f)$ is discrete 1.1. Therefore, $\text{Crit}(\phi_c)$ must be finite. □

Then we give a introduction to the generalization of Andreotti-Frankel theorem.

Definition 5.1. A stratification of a scheme X is a decomposition of X into disjoint Zariski locally closed subset $X = \coprod_{i \in I} X_i$. A finite stratification is a stratification such that I is finite.

Remark 5.1. There are a lot of definitions about stratification with some different conditions. Here we don't add more conditions.

Definition 5.2. A sheaf $\mathcal{F}$ on X is a constructible sheaf if X admits a finite stratification $X = \coprod_{i \in I} X_i$ such that $\mathcal{F}|_{X_i^{\text{an}}}$ is a locally constant sheaf w.r.t. the Euclidean topology.

Example 5.3. If $f : Y \rightarrow X$ is a finite-type morphism of complex algebraic varieties and $\mathcal{F}$ is a constructible sheaf on Y . Then $R^j f_* \mathcal{F}$ are constructible sheaves on X .

Proof. Omit. □

Proposition 5.5. Let $\mathcal{F}$ be a constructible sheaf on irreducible complex variety X . Then there exists an open dense subset U of X such that $\mathcal{F}|_U$ is locally constant.

Proof. Let $X = \coprod_{i \in I} X_i$ be a finite stratification w.r.t. $\mathcal{F}$, then we have

$$X = \overline{\coprod_{i \in I} X_i} = \cup_{i \in I} \overline{X_i}.$$

By irreducibility, there exists i_0 such that $\overline{X_{i_0}} = X$. But we know $X_{i_0} = V \cap F$ is locally closed. Consider

$$X = \overline{X_{i_0}} = \overline{V \cap F} \subseteq \overline{F} = F \Rightarrow F = X.$$

Let $U = X_{i_0}$. □

Theorem 5.6. (Artin-Grothendieck) Let V be an affine irreducible variety of dimension n and $\mathcal{F}$ be a constructible sheaf on V . Then

$$H_{\text{Euclidean}}^i(V, \mathcal{F}) = 0$$

for all $i > n$.

Proof. Consider the finite morphism $i : V \rightarrow \mathbb{C}^n$ given by Noether's normalization theorem. By 2.5 we know that

$$[f_*(\mathcal{F})]_x = \bigoplus_{y \in f^{-1}(x)} \mathcal{F}_y$$

and thus f_* is an exact functor. Then

$$H^i(V, \mathcal{F}) = H^i(\mathbb{C}^n, f_*\mathcal{F}).$$

We now reduce to the case $V = \mathbb{C}^n$. For the rest part, we give a sketch of proof. We induction on n . For $n = 1$, we can assume $X = U \amalg Z$ by 5.5. According to 5.1, $H^2(U, \mathcal{F}|_U) = 0$. Since $Z \subsetneq X$, we know $\dim_{\text{Kru}} Z = 0$, thus $H^2(Z, \mathcal{F}|_Z) = 0$. And we can use some argument to get $H^2(X, \mathcal{F}) = 0$ we omit here. For the case n , first we choose an nonempty algebraic subvariety $Z \subseteq \mathbb{C}^n$ such that $\mathcal{F}|_{\mathbb{C}^n - Z}$ is locally constant by 5.5. And we choose a projection $p : \mathbb{C}^n \rightarrow \mathbb{C}^{n-1}$ such that $p|_Z$ is finite. Then there is a formula

$$(R^i p_* \mathcal{F})_x = H^i(p^{-1}(x), \mathcal{F}|_{p^{-1}(x)}).$$

$\mathcal{F}|_{p^{-1}(x)}$ is a constructible sheaf on $p^{-1}(x) = \mathbb{C}$ and by induction $R^i p_*(\mathcal{F}) = 0$ for all $i > 1$. $p_* \mathcal{F}$ and $R^1 p_* \mathcal{F}$ are both constructible sheaves on $\mathbb{C}^{n-1}$ by 5.3, which says that $H^i(\mathbb{C}^{n-1}, p_* \mathcal{F}) = H^i(\mathbb{C}^{n-1}, R^1 p_* \mathcal{F}) = 0$ for all $i > n - 1$. The conclusion follows by 2.6. For the mentioned formula, use

$$(R^i p_* \mathcal{F})_x = \varinjlim_{\substack{U \text{ open} \\ x \in U}} H^i(f^{-1}(U), \mathcal{F})$$

and 2.3

$$H^i(p^{-1}(x), \mathcal{F}|_{p^{-1}(x)}) = \varinjlim_{\substack{N \text{ open} \\ p^{-1}(x) \subseteq N}} H^i(N, \mathcal{F})$$

Finally use 2.4 to prove the two direct limits are the same. (Here is the most hard part and we omit here, one can consult Lazarsfeld theorem 3.1.13) □

6 Lefschetz Hyperplane Theorem

Theorem 6.1. (Lefschetz hyperplane theorem) Let X be a non-singular irreducible complex projective variety of dimension n , and let D be any effective ample divisor on X . Then

$$H^i(X, D; A) = 0$$

for $i < n$ and any abelian group A . Or equivalently,

$$H^i(X; A) \rightarrow H^i(D; A)$$

is an isomorphism for $i \leq n - 2$ and injective for $i = n - 1$.

Proof. Since D is ample, we can choose a closed embedding $\phi_{|mD|} : X \hookrightarrow \mathbb{P}^r$ such that $H \cap X = mD$. Thus $X - D = X - mD \subseteq \mathbb{P}^r - H = \mathbb{A}^r$ is affine. Thus by 5.1, we know that

$$H_i(X - D; A) = 0 \quad \forall i > n$$

Thus,

$$H^i(X, D; A) \cong H_c^i(X - D; A) \cong H_{2n-i}(X - D; A) = 0$$

by 2.8 and 2.9. □

There is also a version about homopoty groups.

Theorem 6.2. (Lefschetz hyperplane theorem for homotopy groups) Let X be a non-singular irreducible complex projective variety of dimension n , and let D be any effective ample divisor on X . Then

$$\pi_i(X, D) = 0$$

for $i < n$. In particular,

$$\pi_i(D) \rightarrow \pi_i(X)$$

is an isomorphism for $i \leq n - 2$ and surjective for $i = n - 1$. i.e. the inclusion $i : D \rightarrow X$ is a $(n - 1)$ -connected map.

Proof. We choose a projective embedding $X \subseteq \mathbb{P}^r$, a hyperplane $H \subseteq \mathbb{P}^r$ such that $X \cap H = mD$ and thus $V = X - D \subseteq \mathbb{C}^r$ is affine. By 1.3 we know can pick $c \in \mathbb{C}^r$ such that ϕ_c is a Morse function. Now we define

$$f(x) = \begin{cases} \frac{1}{\phi_c(x)} & x \in V = X - D \\ 0 & x \in D \end{cases}$$

Firstly, f is a smooth function. Since, WLOG, we assume that $H = V(z_r)$, $\mathbb{P}^r = \text{Proj}.[z_0, \dots, z_r]$ and $\mathbb{C}^r = \text{Spec } \mathbb{C}[\frac{z_0}{z_r}, \dots, \frac{z_{r-1}}{z_r}]$. Then for any sequence $p_k = (a_i^k) \rightarrow p = (a_i) \in D \subseteq \mathbb{P}^r$, we must have $a_r^k \rightarrow 0$ i.e. $\left\| \left(\frac{a_0^k}{a_r^k}, \dots, \frac{a_{r-1}^k}{a_r^k} \right) \right\| \rightarrow \infty$ in $\mathbb{C}^r$, which implies $\phi_c(p_k) \rightarrow \infty$. Therefore f is continuous on X and smooth on D resp. V , which says f is smooth. Note that on V , f has the same critical points with ϕ_c , but the signs of eigenvalues reversed, since

$$\nabla f(p) = -\frac{1}{\phi_c^2} \nabla \phi_c(p)$$

$$\text{Hess}_p(f) = 2 \frac{1}{\phi_c^3} \nabla \phi_c^T \cdot \nabla \phi_c - \frac{1}{\phi_c^2} \text{Hess}_p(\phi_c) = -\frac{1}{\phi_c^2} \text{Hess}_p(\phi_c)$$

Thus $\text{Index}_p(f) \geq n$ for any $p \in \text{Crit}(f) \cap V$. Now pick any $\varepsilon > 0$, then $D \subseteq X^\varepsilon$ and $X = X^\varepsilon \cup e^{\lambda_1} \cup \dots \cup e^{\lambda_k}$ with $\lambda_i \geq n$. In particular, $\pi_i(X, X^\varepsilon) = 0$ for all $i < n$ by 3.7. Using 3.6 for $D \subseteq X^\varepsilon \subseteq X$ we know

$$\pi_i(X^\varepsilon, D) \rightarrow \pi_i(X, D)$$

is bijective for $i \leq n - 1$. Since X is projective complex variety, by 3.2, it can be triangulated such that D is a simplicial subcomplex. We choose the open star neighborhood U of D , which can deform to D by straight line. Now consider $D \subseteq U \subseteq X^\varepsilon \subseteq X$, by 3.4

$$\begin{array}{ccc} \pi_i(X^\varepsilon, D) & \xrightarrow{\quad} & \pi_i(U, D) \\ & \searrow \cong & \swarrow \\ & \pi_i(X, D) & \end{array}$$

As $i : D \rightarrow U$ is a retraction, in particular a homotopy equivalence, by 3.5, we know $\pi_i(U, D) = 0$ and the conclusion follows. $\square$

Example 6.1. Let X be a nonsingular simply connected projective variety of dimension ≥ 3 , and let $D \subseteq X$ be ample divisor. Then D is simply connected. In particular, hypersurfaces in $\mathbb{P}^n$ is simply connected for $n \geq 3$.

Proof. By above theorem, we know $\pi_1(D) = \pi_1(X) = 0$. $\square$

Corollary 6.3. Let X be a smooth irreducible projective variety of dimension $n \geq 2$ and $D \subseteq X$ an ample divisor. Then D is connected.

Proof. By condition we know $H^0(X, \mathcal{O}_X) = \mathbb{C}$. Together with 4.1, one has

$$H^0(X^{an}; \mathcal{O}_{X^{an}}) = H^0(X, \mathcal{O}_X) = \mathbb{C}$$

which just says that X^{an} is connected. Thus by 6.1, we know that $H^0(D^{an}; \mathbb{Z}) = H^0(X^{an}; \mathbb{Z}) = \mathbb{Z}$, which in turn shows that D is connected. $\square$

Example 6.2. Let X be a non-singular irreducible complex projective variety of dimension n , and let D be any effective ample divisor on X . Then

$$\text{coker}(H^{n-1}(X^{an}; \mathbb{Z}) \rightarrow H^{n-1}(D^{an}; \mathbb{Z}))$$

is torsion-free.

Proof. We use the duality as in the proof of Lefschetz hyperplane theorem

$$H^n(X^{an}, D^{an}; \mathbb{Z}) \cong H_n(X^{an} - D^{an}; \mathbb{Z})$$

which is torsion-free by 5.4 $\square$

Example 6.3. Let X be a smooth projective variety of dimension n , and D a nonsingular ample effective divisor on X , denote by

$$\begin{aligned} r_{p,q} : H^q(X, \Omega_X^p) &\longrightarrow H^q(D, \Omega_D^p) \\ f dz_I \wedge d\bar{z}_J &\longmapsto (f dz_I \wedge d\bar{z}_J)|_D \end{aligned}$$

the natural maps determined by restriction. Then $r_{p,q}$ is isomorphism for $p + q \leq n - 2$ and injective for $p + q = n - 1$

Proof. We use Hodge decomposition 4.3 for X and D , Lefschetz hyperplane theorem 6.1 and note that the induced map keeps the bidegree structure. $\square$

Theorem 6.4. (Kodaira vanishing theorem) Let X be a smooth projective variety of dimension n and D an ample divisor on X , then

$$H^i(X, K_X + D) = 0 \quad \forall i > 0$$

Proof. We omit the covering trick for general case and assume that D is smooth effective. For general case, one need to use the covering of X to reduce to the smooth case. By Grothendieck vanishing theorem and Serre's duality we just need to prove that

$$H^i(X, \mathcal{O}_X(-D)) = 0$$

for all $i < n$. Now consider the short exact sequence

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow \mathcal{O}_D \longrightarrow 0$$

Taking cohomology we get the desired result by 6.3. □

Theorem 6.5. (Nakano vanishing theorem) Let X be a smooth projective variety of dimension n and D an ample divisor on X , then

$$H^q(X, \Omega_X^p \otimes \mathcal{O}_X(D)) = 0 \quad \forall p + q > n$$

Proof. We omit the covering trick also and consider the case of D is smooth ample effective and by Serre's duality we just need to proof that

$$H^q(X, \Omega_X^p(-D)) = 0$$

for any $p + q < n$. Assume that $i : D \rightarrow X$ is the closed embedding, we consider the following commutative diagram given by two exact sequences.

$$\begin{array}{ccccccc}
0 & & & 0 & & & \\
& \searrow & & \downarrow & & & \\
& & \ker f_p & \xleftarrow{\quad} & \Omega_X^p(-D) & \xleftarrow{\quad} & 0 \\
& & \downarrow & \searrow & \downarrow & & \\
& & & & \Omega_X^p & \xrightarrow{f_p} & \\
& & & & \downarrow & & \\
0 & \longrightarrow & i_*\Omega_D^{p-1}(-D) & \longrightarrow & i_*(\Omega_X^p|_D) & \longrightarrow & i_*\Omega_D^p \longrightarrow 0 \\
& & \downarrow & & \downarrow & & \searrow \\
& & 0 & & 0 & & 0
\end{array}$$

The vertical sequence is given by tensoring $- \otimes \Omega_X^p$ with

$$0 \longrightarrow \mathcal{O}_X(-D) \longrightarrow \mathcal{O}_X \longrightarrow i_*\mathcal{O}_D \longrightarrow 0$$

and the horizontal sequence is given by taking p -th wedge (lemma 6.6) on

$$0 \longrightarrow \mathcal{O}_D(-D) \longrightarrow \Omega_X|_D \longrightarrow \Omega_D \longrightarrow 0$$

Taking cohomology on the sequence and using 6.3,

$$0 \longrightarrow \ker f_p \longrightarrow \Omega_X^p \longrightarrow i_*\Omega_D^p \longrightarrow 0$$

we know that $H^q(X, \ker f_p) = 0$ for all $p + q < n$. Now by the lemma 6.7 we know that

$$0 \longrightarrow \Omega_X^p(-D) \longrightarrow \ker f_p \longrightarrow i_*\Omega_D^{p-1}(-D) \longrightarrow 0$$

is exact. Thus by taking cohomology we know

$$H^q(X, \Omega_X^p(-D)) = H^{q-1}(D, \Omega_D^{p-1}(-D))$$

Therefore by induction on n we reduce to the case $n = 1$. i.e. $H^0(X, \mathcal{O}_X(-D)) = 0$, which is given by 6.4. $\square$

Lemma 6.6. Let

$$0 \longrightarrow \mathcal{L} \longrightarrow \mathcal{F} \longrightarrow \mathcal{E} \longrightarrow 0$$

be an exact sequence of locally free sheaf with $\mathcal{L}$ invertible, then

$$0 \longrightarrow \mathcal{L} \otimes \wedge^{p-1} \mathcal{E} \longrightarrow \wedge^p \mathcal{F} \longrightarrow \wedge^p \mathcal{E} \longrightarrow 0$$

is exact.

Proof. Check on basis. $\square$

Lemma 6.7. Consider the following commutative diagram with the vertical and horizontal sequences exact.

$$\begin{array}{ccccccc}
 & & 0 & & 0 & & \\
 & \searrow & & \downarrow & & \downarrow & \\
 & & \ker f & \xleftarrow{\quad} & E & \xleftarrow{\quad} & 0 \\
 & & \downarrow & & \downarrow & & \\
 & & & & D & \searrow f & \\
 & & & & \downarrow & & \\
 0 & \longrightarrow & A & \longrightarrow & B & \longrightarrow & C \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \searrow \\
 & & 0 & & 0 & & 0
 \end{array}$$

Then we have an exact sequence

$$0 \longrightarrow E \longrightarrow \ker f \longrightarrow A \longrightarrow 0$$

Proof. Diagram chasing. $\square$

Example 6.4. (Lefschetz theorem for Picard groups) Let X be a smooth projective variety of dimension ≥ 4 , and $D \subseteq X$ a reduced effective ample divisor. Then the map

$$\text{Pic}(X) \rightarrow \text{Pic}(D)$$

is an isomorphism.

Proof. We consider the exponential sequence

$$\begin{array}{ccccccc}
 0 & \longrightarrow & \mathbb{Z}_X & \longrightarrow & \mathcal{O}_{X^{an}} & \xrightarrow{\exp} & \mathcal{O}_{X^{an}}^* \longrightarrow 0 \\
 & & \downarrow & & \downarrow & & \downarrow \\
 0 & \longrightarrow & \mathbb{Z}_D & \longrightarrow & \mathcal{O}_{D^{an}} & \xrightarrow{\exp} & \mathcal{O}_{D^{an}}^* \longrightarrow 0
 \end{array}$$

taking cohomology, using $H^1(X^{an}, \mathcal{O}_{X^{an}}^*) = \text{Pic}(X^{an}) = \text{Pic}(X)$ (see 4.2) and $H^i(X; \mathbb{Z}) = H^i(X; \mathbb{Z}_X)$ (see 2.7), we can get

$$\begin{array}{ccccccccc}
 H^1(X; \mathbb{Z}) & \longrightarrow & H^1(X, \mathcal{O}_X) & \longrightarrow & \text{Pic}(X) & \longrightarrow & H^2(X; \mathbb{Z}) & \longrightarrow & H^2(X, \mathcal{O}_X) \\
 \downarrow 6.1 & & \downarrow 6.4 & & \downarrow & & \downarrow 6.4 & & \downarrow 6.1 \\
 H^1(D; \mathbb{Z}) & \longrightarrow & H^1(D, \mathcal{O}_D) & \longrightarrow & \text{Pic}(D) & \longrightarrow & H^2(D; \mathbb{Z}) & \longrightarrow & H^2(D, \mathcal{O}_D)
 \end{array}$$

Thus $\text{Pic}(X) \cong \text{Pic}(D)$ by five lemma. $\square$

Example 6.5. Assume $n + 1 \geq 5$ and let $F \in \mathbb{C}[x_0, \dots, x_n]$ be a non-zero homogeneous polynomial of degree d such that $D = V(F)$ is non-singular. Then

$$R = \frac{\mathbb{C}[x_0, \dots, x_n]}{(F)}$$

is UFD.

Proof. We use the criteria that a Noetherian domain R is UFD iff R is normal and $\text{Cl}(\text{Spec}(R)) = 0$. (See Hartshorne chapter 2 section 6 proposition 6.2). D is non-singular, so R is normal by Serre's criteria. For the divisor class group, we consider $C = \text{Spec}(R)$, $X = \text{Proj } R$, $E = \text{Spec}_X \text{Sym}^\bullet \mathcal{O}(1)$ and $p : E \rightarrow X$ be the projection. Let Z be the zero section of p which locally defined by $t = 0$. Then we have $\text{Cl}(E) \cong \text{Cl}(X)$ by Hartshorne chapter 2 section 6 proposition 6.6 or exercise 6.3. By local check, one can identify $E - Z$ with $C - \{R_+\}$. Finally we use the exact sequence

$$\mathbb{Z} \longrightarrow \text{Cl}(E) \longrightarrow \text{Cl}(E - Z) \longrightarrow 0$$

we have

$$\begin{aligned} \text{Cl}(C) &\cong \text{Cl}(C - \{R_+\}) \cong \text{Cl}(E - Z) \\ &\cong \text{Cl}(E)/\mathbb{Z} \cdot [Z] \cong \text{Cl}(X)/\mathbb{Z} \cdot [\mathcal{O}_X(-1)] \\ &\cong \text{Pic}(X)/\mathbb{Z} \cdot [\mathcal{O}_X(-1)] \\ &\cong \text{Pic}(\mathbb{P}^n)/\mathbb{Z} \cdot [\mathcal{O}_{\mathbb{P}^n}(-1)] \text{ (use 6.4)} \\ &= 0 \end{aligned}$$

□

Lemma 6.8. Let X be a smooth projective complex variety of dimension n , then

$$N^1(X) = H^2(X; \mathbb{Z})_{\text{t.f.}} \cap H^{1,1}(X, \mathbb{C})$$

Proof. See Lazarsfeld remark 1.1.21.

□

Example 6.6. (Isomorphism of N^1) Let X be a smooth projective variety of dimension ≥ 4 and $D \subseteq X$ a smooth ample effective divisor. Then

$$N^1(X)_{\mathbb{R}} \cong N^1(D)_{\mathbb{R}}$$

Proof. We use 6.8, 6.3 and 6.1.

□

Example 6.7. (Non-isomorphism of Nef) Let $X = \mathbb{P}^1 \times \mathbb{P}^{n-1}$ and $D \subseteq X$ be a general divisor of type (a, b) with $a \geq n + 1, b > 0$ and $n \geq 4$. Assume

$$\mathcal{L}_k = \text{pr}_1^* \mathcal{O}_{\mathbb{P}^1}(-1) \otimes \text{pr}_2^* \mathcal{O}_{\mathbb{P}^{n-1}}(k),$$

then $\mathcal{L}_k|_D$ is ample for $k \gg 0$ but $\mathcal{L}_k$ on X is not nef nor big. Thus $\text{Nef}(X)|_D$ cannot be identify with $\text{Nef}(D)$ under restriction.

Proof. Let $D = V(F) \subseteq \mathbb{P}^1 \times \mathbb{P}^{n-1}$ be a smooth effective ample divisor defined by bihomogeneous polynomial F of bidegree (a, b) with $a \geq n$. Then the projection $\varphi : D \hookrightarrow \mathbb{P}^1 \times \mathbb{P}^{n-1} \rightarrow \mathbb{P}^{n-1}$ is finite since it is proper

and quasi-finite. Here properness is obvious. For quasi-finite property, given a point $y = [y_0 : \cdots : y_{n-1}]$ of $\mathbb{P}^{n-1}$, the polynomial $F(x_0, x_1; y_0, \cdots, y_{n-1}) = 0$ has only finite solution if $g_y(x) := F(x, y) \neq 0$. The condition $g_y \neq 0$ is ensured by the condition $a \geq n$. Assume $g_y(x) = \sum_{i=0}^a F_i(y) x_0^i x_1^{a-i}$, then $g_y(x) \neq 0$ is equivalent to say that $\emptyset = \bigcap_{i=0}^a V(F_i(y)) \subseteq \mathbb{P}^{n-1}$, which can be done by pick a general D (s.t. $F_0, \cdots, F_{n-1}$ is a regular sequence and pick F_n not passing the finite points $\bigcap_{i=0}^{n-1} V(F_i(y))$.) Thus we know $\varphi^* \mathcal{O}_{\mathbb{P}^{n-1}}(1)$ is ample. Now let $\mathcal{L}_k = \text{pr}_1^* \mathcal{O}_{\mathbb{P}^1}(-1) \otimes \text{pr}_2^* \mathcal{O}_{\mathbb{P}^{n-1}}(k)$. Then $\mathcal{L}_k|_D$ is ample for $k \gg 0$ since $\mathcal{L}_k|_D = \text{pr}_1^* \mathcal{O}_{\mathbb{P}^1}(-1)|_D \otimes \varphi^* \mathcal{O}_{\mathbb{P}^{n-1}}(k)$ and $\varphi^* \mathcal{O}_{\mathbb{P}^{n-1}}(1)$ is ample. But $\mathcal{L}_k$ is not nef nor big. Let $C = \mathbb{P}^1 \times \{p\}$, then $(\mathcal{L}_k \cdot C) = -1 + 0 = -1$ and by Kunneth formula

$$h^0(\mathbb{P}^1 \times \mathbb{P}^{n-1}, \mathcal{L}_k^{\otimes m}) = H^0(\mathbb{P}^1, \mathcal{O}_{\mathbb{P}^{n-1}}(-1)) \otimes H^0(\mathbb{P}^{n-1}, \mathcal{O}_{\mathbb{P}^{n-1}}(mk)) = \binom{n-1+mk}{n-1}$$

which shows that $\text{vol}(\mathcal{L}_k) = \limsup_{m \rightarrow \infty} \frac{h^0(\mathbb{P}^1 \times \mathbb{P}^{n-1}, \mathcal{L}_k^{\otimes m})}{\frac{m^n}{n!}} = \limsup_{m \rightarrow \infty} \frac{1}{m} = 0$. $\square$

Example 6.8. (Complete intersection) Suppose $Y \subseteq \mathbb{P}^n$ is the transversal complete intersection of hypersurfaces $D_1, \cdots, D_e \subseteq \mathbb{P}^r$ of degree $d_1 \geq \cdots \geq d_e$. Then

$$H^i(\mathbb{P}^n; Y; \mathbb{Z}) = 0$$

for $i \leq n - e$.

Proof. We modify the divisor such that $Y_d := D_1 \cap \cdots \cap D_d$ is smooth. Then by Lefschetz hyperplane theorem we know that

$$H^i(Y_d, Y_{d+1}; \mathbb{Z}) = 0 \quad \forall i < n - d$$

Use induction and long exact sequence of cohomology one can get the desired result. $\square$

Example 6.9. (Counterexample for Lefschetz hyperplane theorem) Let $\mathbb{P}_{\mathbb{C}}^3 = \text{Proj.} \mathbb{C}[x_0, x_1, x_2, x_3]$, $\mathbb{P}_{\mathbb{C}}^{18} = \text{Proj.} \mathbb{C}[z_0, z_1, \cdots, z_{17}, z_{18}]$ and

$$V = \text{span}_{\mathbb{C}} \{x_0^i x_1^j x_2^k x_3^l, x_0^3 + x_1^3\}_{\substack{(i,j,k,l) \neq (3,0,0,0), (0,3,0,0) \\ i+j+k+l=3}}$$

be a free linear system (directly check that they have no common zeros) and $\phi|_V : \mathbb{P}^3 \longrightarrow \mathbb{P}^{18}$ be the morphism defined by V . Assume

$$X = \text{im}^{\text{Sch}} \phi|_V = \overline{\text{im}^{\text{Set}} \phi|_V} = \text{im}^{\text{Set}} \phi|_V$$

as sets since $\phi|_V$ is proper and hence closed map. We equip X with reduced induced structure. For convenience, let $p = [1 : 0 : 0 : 0]$ and $q = [0 : 1 : 0 : 0]$ and $o = [1 : 0 : \cdots : 0]$. Pick a divisor $\tilde{D} = V(z_{x_0^3+x_1^3} + z_{x_2^3} + z_{x_3^3}) \subseteq \mathbb{P}^{18}$ (here $z_{x_0^3+x_1^3}$ denote the corresponding coordinate of $x_0^3 + x_1^3$ in $\mathbb{P}^{18}$) and let $D = \tilde{D}|_X$. Then $H^1(X, D; \mathbb{Z}) \neq 0$.

Proof. As topological space, $X^{an} \cong \mathbb{P}^3 / \{p, q\} \sim \mathbb{P}^3 \vee S^1$, thus

$$H^1(X; \mathbb{Z}) = H^1(\mathbb{P}^3; \mathbb{Z}) \oplus H^1(S^1; \mathbb{Z}) = \mathbb{Z}$$

For $H^1(D; \mathbb{Z})$ we consider the induced morphism $\phi|_V : \mathbb{P}^3 \rightarrow X$. Note that $\mathbb{P}^3 - \{p, q\} = D_+(x_2) \cup D_+(x_3) \cup D_+(x_0 x_1)$ and one can check that this is isomorphism on each patch. Since $o \notin D$, under this isomorphism, $E := \phi|_V^*(D) = V(x_0^3 + x_1^3 + x_2^3 + x_3^3) \subseteq \mathbb{P}^3$ is a smooth effective ample divisor. Then we can use Lefschetz hyperplane theorem 6.1

$$H^1(D; \mathbb{Z}) = H^1(\mathbb{P}^3; \mathbb{Z}) = 0$$

Finally we use long exact sequence and $H^0(X; \mathbb{Z}) \cong H^0(D; \mathbb{Z})$ to deduce $H^1(X, D; \mathbb{Z}) \neq 0$. $\square$